

Friday warm-up: Forces I

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1 Memory Bank

1. $\vec{F}_{\text{net}} = m\vec{a}$... Newton's Second Law, relating net force, mass, and acceleration.
2. Units of force: $1 \text{ N} = 1 \text{ kg m s}^{-2}$... The definition of a Newton of force, like 1 pound.
3. $g = 9.81 \text{ m s}^{-2}$... The gravitational acceleration near the Earth's surface.

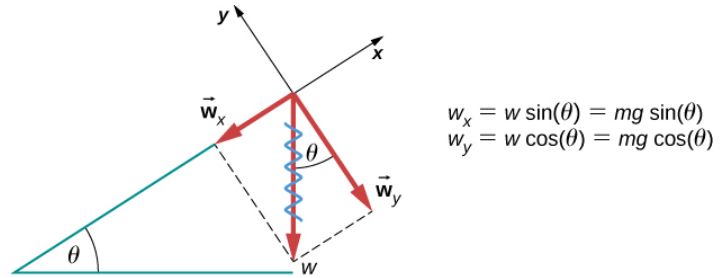


Figure 1: Forces on an object on an incline.

2 Chapter 5 - Forces

1. Prove that the two θ in Fig. 1 are equal, using geometric techniques and the properties of triangles.
2. (a) Using Fig. 1, show that the magnitude of the weight force is mg , given the components w_x and w_y . (b) Show that, in the limit $\theta \rightarrow 90$ degrees, w_x approaches the full weight of the system. (c) Show that, in the limit $\theta \rightarrow 0$ degrees, w_y approaches the full weight of the system.
3. Suppose an object with mass 40 kg is sliding down an incline (Fig. 1). In Figure 1, the weight force $\vec{w}$ is broken into components $\vec{w}_x$ and $\vec{w}_y$. (a) If $\theta = 30$ degrees, calculate the magnitudes w_x and w_y . (b) What is the net force on the system, assuming there is no friction? (c) If the system slides 0.45 meters starting from rest, what is the final speed?
4. Suppose a 60 kg skier is sliding down a slope with $\theta = 15$ degrees. Assume a frictional force of 40 N opposes the motion. (a) What is the net force on the skier? (b) What is the acceleration of the skier? (c) Starting with an initial velocity of 5 m s^{-1} , how far does the skier travel in 30 seconds?